My portfolio

Nathan Verdier

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PROFILE

Cybersecurity engineering student at ESIEA, apprentice at IMAAAGERY (ISO 27001 ISMS, machine learning). EBIOS Risk Manager certified, 4th at the CND 2025 hackathon, top 2 % on TryHackMe.

PROFESSIONAL EXPERIENCE

IMAAAGERY — Cybersecurity (ISO 27001 ISMS) & Machine Learning (apprenticeship) Apr. 2025 – present
Security and ML engineer *Clermont-Ferrand, France*

- **ISO 27001:2022 ISMS**: end-to-end project; Annex A controls A1 to A45, information security policy (PSSI) and the 13 ANSSI recommendations for small businesses (guide "La cybersécurité pour les TPE/PME en 13 questions").
- **Architecture**: moved a self-hosted infrastructure to a Microsoft cloud stack; UniFi network, VLANs, WPA3-Enterprise/RADIUS.
- **Compliance & awareness**: employee training, CNIL obligations, GDPR, ISO 27001; business continuity plan (BCP) and disaster recovery plan (DRP).
- **Backup**: incremental encrypted backup on a Synology NAS, replicated to the cloud.
- **Machine learning**: Projet Lumière (smart photo capture and "shade finder" returning the true skin tone); SunDamage pipeline (CLAHE, MediaPipe, Daubechies wavelets, Otsu thresholding adapted to skin phototype).

IMAAAGERY — Web, Machine Learning & Deep Learning (apprenticeship) 2023 – 2024
Segmentation, classification and clustering (PyTorch, YOLOv8, KNN); web tools in React JS and Radix UI.

Université du Québec à Chicoutimi (UQAC) — Video game development (internship, 10 weeks) Apr. – June 2023
Unreal Engine, C++ and Perforce version control, in the computer science lab (Canada).

EDUCATION

ESIEA — Engineering degree, cybersecurity major (5th year) 2025 – present
Security methodology (ENT5003), local network security, firewalls and proxies, auditing, forensics; Qoracle research project.

Université Clermont Auvergne (EUPI) — M1 Master's in Embedded Systems, Signal, Image and Sound Processing 2024 – 2025

Université Clermont Auvergne (IUT) — Bachelor of Technology in Computer Science 2021 – 2024
"Numérique responsable" (responsible digital) award at the 24h pour entreprendre.

UQAC (Canada) — Short computer science programme for exchange students Jan. – June 2023

CERTIFICATIONS & AWARDS

- **EBIOS Risk Manager (EBIOS RM) certification** — cybersecurity risk analysis.
- **SecNumacadémie MOOC (ANSSI)** and **GDPR MOOC (CNIL)** — completed and validated.
- **AWS Academy Cloud Foundations** and **AWS Academy Cloud Security Foundations**.
- **CND 2025 hackathon, Mont-Valérien** — 4th place • **"Numérique responsable" award** at the 24h pour entreprendre.

HANDS-ON TRAINING (DAILY FOR 1 YEAR)

- **Hack The Box (Academy and Labs)**: Junior Cybersecurity Analyst path at 89 % (16 of 20 modules), preparing the CJCA certification; level 42, 26 consecutive weeks of activity.
- **TryHackMe**: top 2 % of the ranking, Gold league; Pre Security and Cyber Security 101 paths completed; AI Security in progress (59 %), SOC Level 1 and Junior Penetration Tester started; 268-day streak.
- **Root-Me**: starting point in web security, 24 challenges solved (HTTP, file upload, open redirect, IP filtering bypass).

PROJECTS

Qoracle — Quantum oracle synthesis (ESIEA research, supervised by Arnaud Valence) 2026
Effect of Boolean circuit structure on quantum gate cost; pipeline and Python package `qoracle` with a web interface, covering three syntheses (Boolean to reversible, reversible to elementary gates, Boolean to elementary gates).

- **SyncUp Fitness** (Oct. 2024 – Mar. 2025): Python, Flutter, Dart, MediaPipe, OpenCV, Raspberry Pi, IMX477 camera.
- **DouShouQi_App** (May – June 2024): Swift, iOS, unit and integration tests.
- **SecuMeltdown** (Sept. – Oct. 2023): Python, Bash, Debian.

SKILLS

Security & compliance: ISO 27001:2022, EBIOS RM, PSSI, ANSSI recommendations for small businesses, risk analysis, CNIL obligations, SQL injection, XSS, cryptography, buffer overflow
Network & infrastructure: Wi-Fi, NAT, routing, UniFi, VLANs, WPA3-Enterprise/RADIUS, WireGuard, Entra ID, Intune, Defender for Business
DevSecOps & ML: CI/CD, SonarQube, Trivy, Burp Suite; PyTorch, YOLOv8, KNN, OpenCV, MediaPipe
Languages: Python, C, C++, C#, Java, Bash, .NET, VHDL, PHP, Swift, Kotlin, Dart
Web, mobile & databases: React JS, React Native, Flutter, HTML, CSS; Oracle SQL, PostgreSQL, MongoDB
Systems & embedded: Linux (Debian 12, Parrot OS), Windows 11; Arduino, Raspberry Pi 4, ESP32, FPGA (MAX10)

LANGUAGES & INTERESTS

Languages: French (native), English (B2)
Interests: HackTheBox, hardware hacking, weight training (10 years), tennis (3 years), guitar